

```
def bellman_ford(vertices, edges, source):
```

```
    distance = [9999] * vertices
```

```
    distance[source] = 0
```

```
    for i in range(vertices - 1):
```

```
        for u, v, weight in edges:
```

```
            if distance[u] != 9999 and \
                distance[u] + weight < distance[v]:
```

```
                distance[v] = distance[u] + weight
```

```
    for u, v, weight in edges:
```

```
        if distance[u] != 9999 and \
            distance[u] + weight < distance[v]:
```

```
            print("Negative weight cycle exists")
            return
```

```
    print("Shortest distances from source", source)
```

```
    for i in range(vertices):
```

```
        print("Vertex", i, "=", distance[i])
```

```
vertices = 5
```

```
edges = [
```

```
    (0, 1, 6),
    (0, 2, 7),
    (1, 2, 8),
    (1, 3, 5),
    (1, 4, -4),
    (2, 3, -3),
    (2, 4, 9),
    (3, 1, -2),
    (4, 0, 2),
    (4, 3, 7)
```

```
]
```

```
source = 0
```

```
bellman_ford(vertices, edges, source)
```

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2025)
on win32
```

```
>>> Enter "help" below or click "Help" above for more.
```

```
==== RESTART: C:/Users/ROOPA SAI DIVYA/OneDrive/Desktop/Python/Python3/Python3.14.3/Python3.14.3>
```

```
Shortest distances from source 0
```

```
Vertex 0 = 0
```

```
Vertex 1 = 2
```

```
Vertex 2 = 7
```

```
Vertex 3 = 4
```

```
Vertex 4 = -2
```

```
>>>
```